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Questions

Q1.

Errors in DNA replication can give rise to mutations.

The diagram shows the bases in a length of DNA.

Length of DNA	A	T	G	C	T	C	A	T	T	T	A	C	C	A	T	C	G	A
Base number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

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The table shows the genetic code for the amino acids.

Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid
AAA AAG	Lysine	CAA CAG	Glutamine	GAA GAG	Glutamic acid	TAC TAT	Tyrosine
AAC AAT	Asparagine	CAT CAC	Histidine	GAC GAT	Aspartic acid	TCA TCC TCG TCT	Serine
ACA ACC ACG ACT	Threonine	CCA CCC CCG CCT	Proline	GCA GCC GCG GCT	Alanine	TGG	Tryptophan
AGA AGG	Arginine	CGA CGC CGG CGT	Arginine	GGA GGC GGG GGT	Glycine	TGC TGT	Cysteine
AGC AGT	Serine	CTA CTC CTG CTT	Leucine	GTA GTC GTG GTT	Valine	TTA TTG	Leucine
ATA ATC ATT	Isoleucine					TTC TTT	Phenylalanine
ATG	Methionine						

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The genetic codes TAA, TAG and TGA are stop codons.

(a) State the sequence of the first four amino acids coded for by this length of DNA.

(1)

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(b) A change in a single base can cause a change in the amino acid sequence produced in protein synthesis.

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(i) Name the type of each mutation described below.

(2)

Base number 3 becomes cytosine (C)

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Base number 6 becomes number 5 in the sequence

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Base number 9 becomes number 10 in the sequence

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*(ii) Explain the possible effects of these three types of mutation on the amino acid sequence coded for by this length of DNA.

Use the information in the table to support your answer.

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(Total for question = 9 marks)

Q2.

Cystic fibrosis is an inherited recessive disease caused by mutations in a gene on chromosome 7.

(a) Give the meaning of the term **gene**.

(1)

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(b) Explain how mutations result in cystic fibrosis.

(3)

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(c) Population carrier screening (PCS) is one type of genetic screening.

This involves screening people who want a child to see if they are carriers of genetic disorders.

In one country, the number of babies born with cystic fibrosis went down following the introduction of PCS.

Suggest why the number of babies born with cystic fibrosis went down.

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(Total for question = 7 marks)

Q3.

Genetic screening can be used to test for aneuploidy.

Aneuploidy is the presence of an abnormal number of chromosomes in a cell.

Aneuploidy can affect the miscarriage rate of implanted embryos.

Following screening, only embryos with the correct number of chromosomes are implanted into the female.

The table shows the miscarriage rate of two groups of implanted embryos:

- embryos not screened for aneuploidy
- embryos screened and shown not to have aneuploidy.

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Age range of women at implantation / years	Miscarriage rate (%)	
	Implanted embryos not screened for aneuploidy	Implanted embryos screened and shown not to have aneuploidy
<35	12.0	11.2
35 to 37	16.8	13.0
38 to 40	25.0	13.6
41 to 42	37.9	16.3
>42	58.8	17.2

(a) Explain how this data shows that there is a correlation between the age of the women and the miscarriage rate.

(2)

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(b) (i) Explain the conclusions that can be made from these data about the causes of miscarriage.

(2)

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(ii) Explain why conclusions made using these data may not be valid.

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(c) Discuss the implications of screening embryos for aneuploidy before implantation.

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(Total for question = 10 marks)

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Q4.

A gene contains the genetic code for the sequence of amino acids in a polypeptide chain.

The table shows the genetic codes found in DNA.

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Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid
AAA AAG	Lysine	CAA CAG	Glutamine	GAA GAG	Glutamic acid	TAC TAT	Tyrosine
AAC AAT	Asparagine	CAT CAC	Histidine	GAC GAT	Aspartate	TCA TCC TCG TCT	Serine
ACA ACC ACG ACT	Threonine	CCA CCC CCG CCT	Proline	GCA GCC GCG GCT	Alanine	TGG	Tryptophan
AGA AGG	Arginine	CGA CGC CGG CGT	Arginine	GGA GGC GGG GGT	Glycine	TGC TGT	Cysteine
AGC AGT	Serine	CTA CTC CTG CTT	Leucine	GTA GTC GTG GTT	Valine	TTA TTG	Leucine
ATA ATC ATT	Isoleucine					TTC TTT	Phenylalanine
ATG	Methionine						

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The genetic codes TAA, TAG and TGA are stop codons, that do not code for amino acids.

*(a) Explain the nature of the genetic code.

Use information in the table to support your answer.

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(b) The diagram shows the sequence of nucleotide bases in part of a DNA template (antisense) strand.

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Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Base	A	T	G	G	C	T	T	G	C	C	C	G	A	T	C	C	T	A

(i) Give the sequence of amino acids that is coded for by these bases.

(1)

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(ii) Explain the possible effects on a protein if there is a substitution mutation in the 9th base in this DNA strand.

Use information in the table to support your answer.

(5)

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(Total for question = 12 marks)

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Q5.

Red-green colour blindness is a common trait in humans.

(a) The gene for red-green colour blindness is located on the X chromosome.

State what is meant by the term **gene**.

(1)

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(b) Describe how the two strands of DNA forming the double helix in a gene are held together.

(2)

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(c) Explain why each codon for the DNA genetic code must contain at least three bases.

(3)

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(d) A red-green colour blind father and an unaffected heterozygous mother had a child.

Determine the probability of this child being red-green colour blind.

Use a genetic diagram to support your answer.

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(3)

Probability .....

(Total for question = 9 marks)

Q6.

Haemophilia A is an inherited genetic disorder.

Haemophilia A results in the blood not being able to clot.

(a) The table gives some information about blood clotting components.

Complete the table by filling in the empty boxes with either the name of the component, the role of the component or the solubility of the component.

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Name	Role	Solubility
thromboplastin	.....	soluble
.....	catalyses the conversion of fibrinogen into fibrin	.....
fibrin	1	.....

(b) Haemophilia A is inherited in a similar way to red-green colour blindness.

Explain why more males than females are affected with haemophilia A.

(3)

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(Total for question = 9 marks)

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Q7.

Rabbits can have brown fur or white fur.

A heterozygous rabbit will have brown fur.

(a) Draw a genetic diagram to show the genotypes and corresponding phenotypes of the baby rabbits produced if two heterozygous rabbits were bred together.

(3)

(b) A number of heterozygous pairs of rabbits were bred together and produced 284 baby rabbits.

Calculate the expected number of homozygous brown rabbits, heterozygous brown rabbits and white rabbits produced.

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(3)

Number of homozygous brown rabbits .....

Number of heterozygous brown rabbits .....

Number of white rabbits .....

(Total for question = 6 marks)

Q8.

Most Bengal tigers are orange with black stripes but there is a very small number of Bengal tigers that are white with black stripes.

The photograph shows a white Bengal tiger with black stripes.

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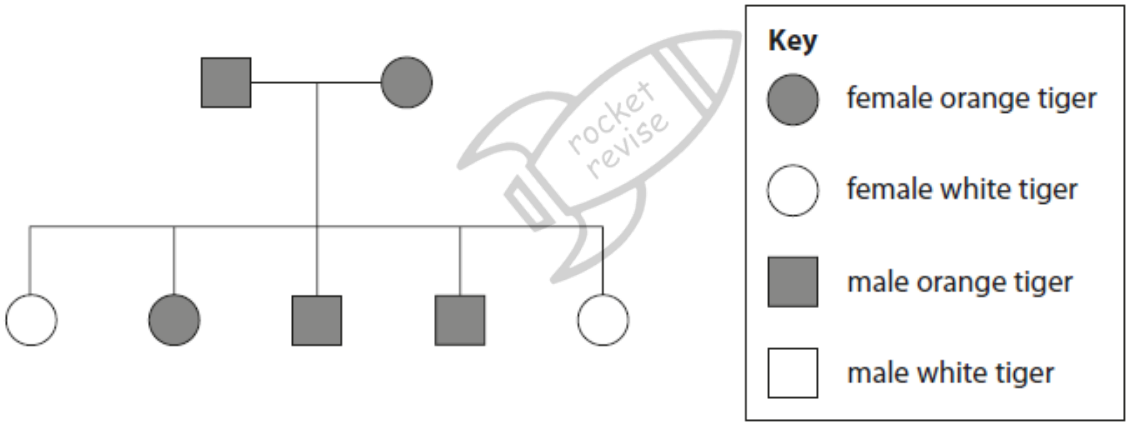


(Source: Caroline Wilcox)

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White tiger offspring are produced by two Bengal tigers that each carry at least one recessive allele for a gene affecting coat colour.

The pedigree diagram shows the phenotypes in one family of tigers, bred in captivity.



(a) The phenotype is affected by the genotype.

State what is meant by the term **genotype**.

(1)

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(b) State the probability that the next tiger born to these two parents will be female.

(1)

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(c) Determine the expected phenotypic ratio of orange tigers to white tigers born to the parents shown in this pedigree diagram.

Use a genetic diagram to support your answer.

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(3)

Answer .....

(d) The incidence of white tigers in the wild is 1 in 10 000 Bengal tigers.

There are approximately 6000 Bengal tigers in captivity, 200 of which are white.

Calculate the incidence of white tigers in captivity.

(1)

Answer .....

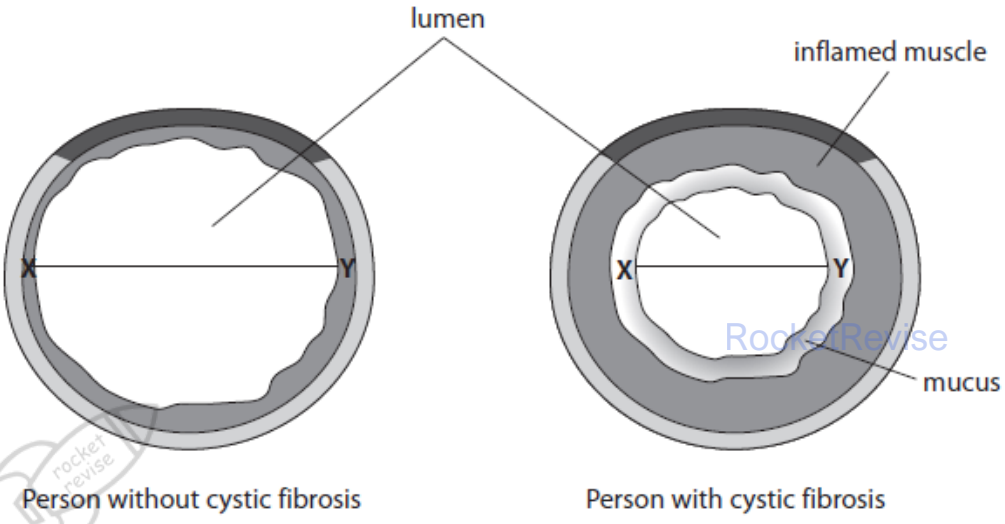
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(Total for question = 6 marks)

Q9.

Cystic fibrosis is an inherited disorder.

(a) The diagrams show a cross section through a trachea from a person without cystic fibrosis and from a person with cystic fibrosis.



(i) Calculate the percentage decrease in the diameter of the lumen of the trachea from a person with cystic fibrosis.

Take your measurements between **X** and **Y** on each diagram.

(2)

Answer .....

* (ii) Explain why cystic fibrosis causes the differences shown in the diagrams.

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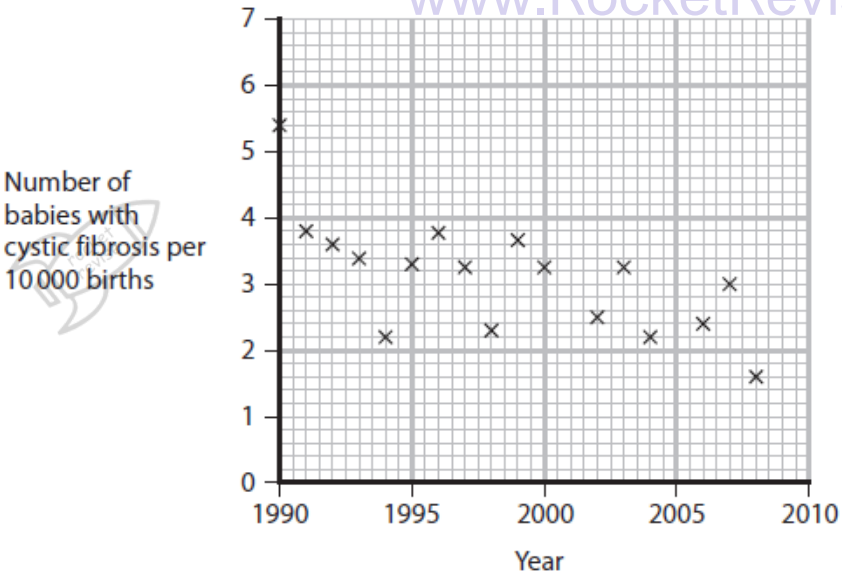
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(b) The graph shows the number of babies with cystic fibrosis per 10 000 births in one country, from 1990 to 2008.

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(i) Explain how the identification of carriers, using genetic screening, could have contributed to the downward trend shown in this graph.

(3)

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(ii) Describe the ethical issues relating to genetic screening.

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(Total for question = 14 marks)

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Q10.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

People with cystic fibrosis produce very thick, sticky mucus.

Cystic fibrosis is caused by mutations in a gene coding for the CFTR protein.

(a) Explain why a mutation in this gene results in the production of very thick, sticky mucus.

(3)

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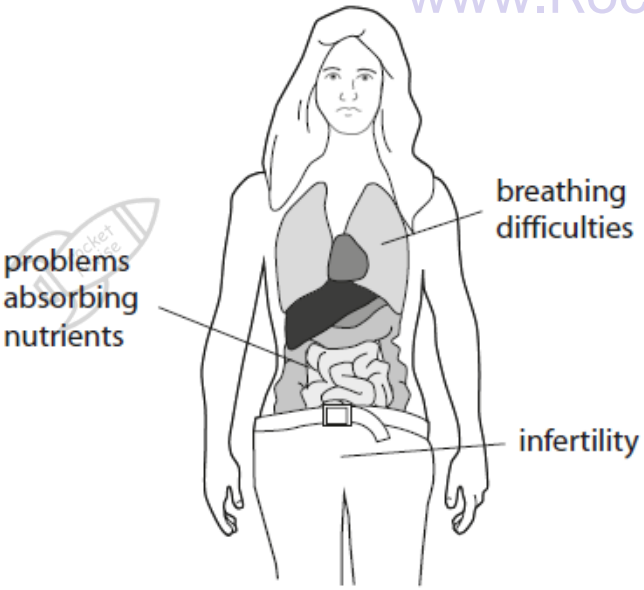
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(b) The diagram shows some health problems associated with cystic fibrosis, in a female.

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Explain why very thick, sticky mucus results in these health problems.

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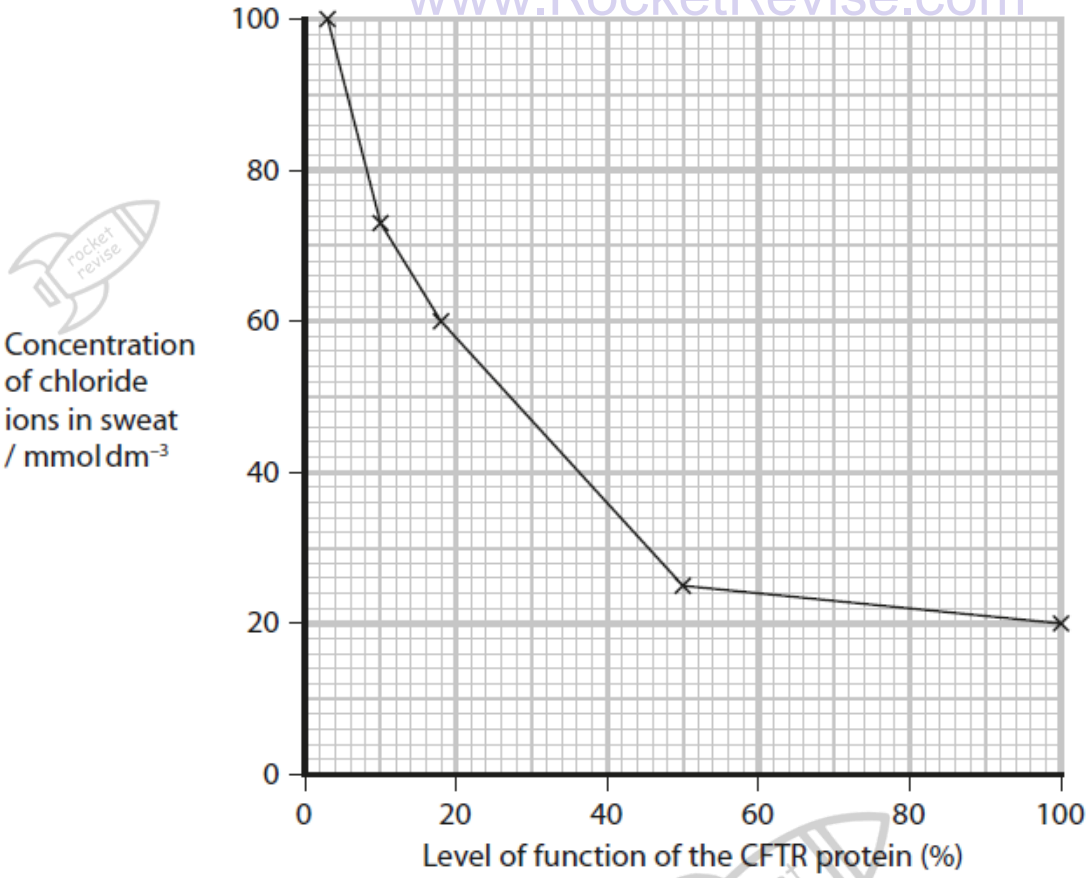
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(c) The graph shows the correlation between the concentration of chloride ions in sweat and the level of function of the CFTR protein.

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Individuals diagnosed with cystic fibrosis have a level of function of the CFTR protein of 18% or less.

(i) Which is the change in concentration of chloride ions in the sweat of an individual when the level of function of CFTR protein decreased from 100% to 18%?

(1)

☒ **A** 15 mmol dm^{-3}

☒ **B** 35 mmol dm^{-3}

☒ **C** 40 mmol dm^{-3}

☒ **D** 80 mmol dm^{-3}

(ii) Cystic fibrosis results from different mutations in the CFTR gene.

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Explain how the graph provides evidence that cystic fibrosis results from different mutations in the CFTR gene.

(2)

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(Total for question = 11 marks)

Q11.

A number of diseases are associated with lifestyle risk factors.

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Some of these risk factors cause mutations.

(a) Mutations can give rise to cancer.

State the meaning of the term **mutation**.

(1)

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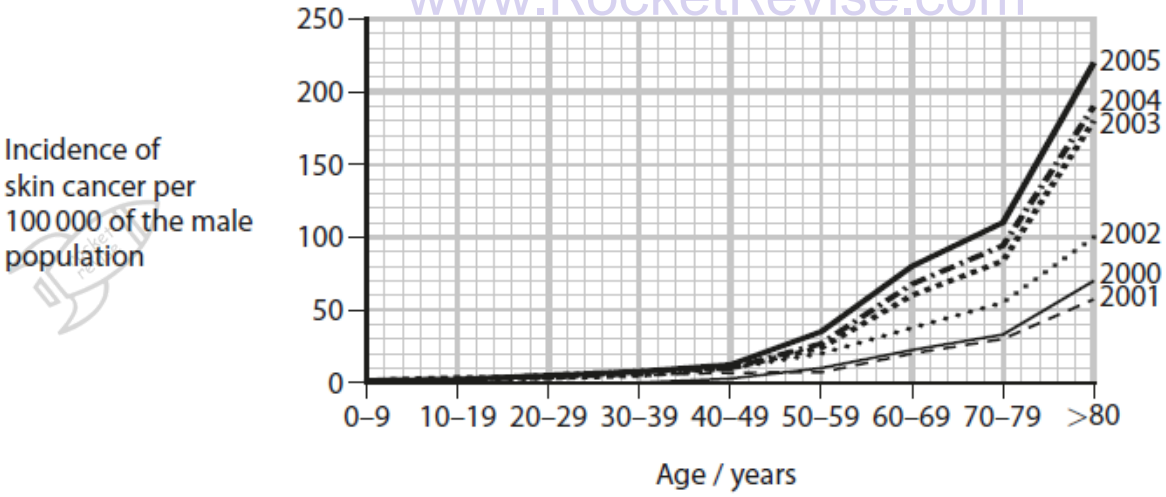
(b) Exposure to ultraviolet light is associated with the development of skin cancer.

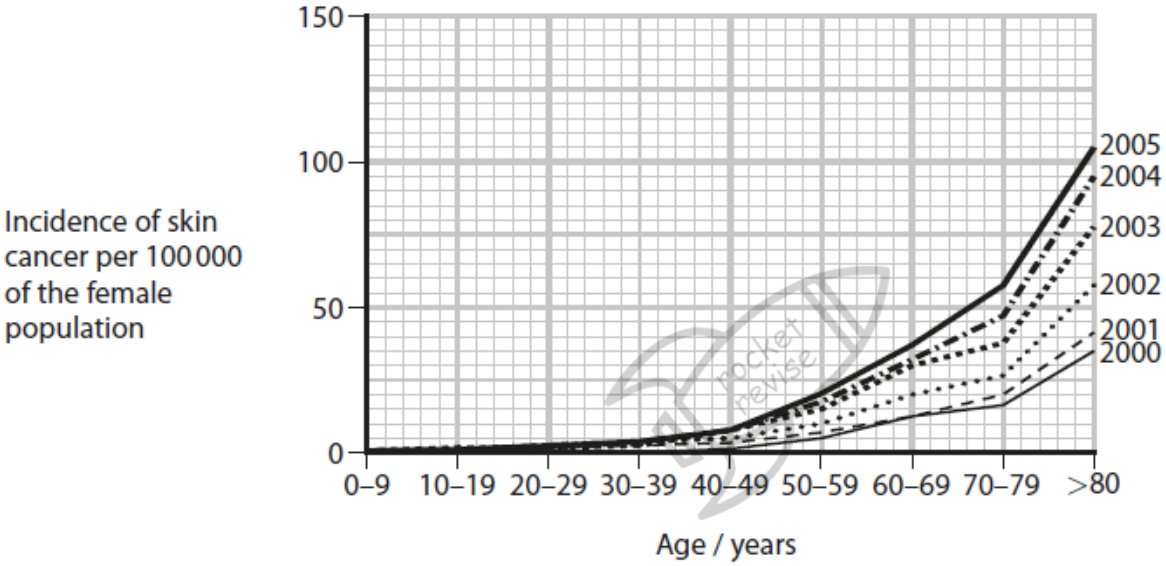
The graphs show the incidence of skin cancer in males and females in one country in the Far East, from 2000 to 2005.

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(i) The graphs show some correlations.

State the meaning of the term **correlation**.

(1)

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(ii) Describe the correlations shown by these graphs.

(3)

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(iii) Suggest a reason for each of the correlations shown by these graphs.

(3)

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*(c) The table shows some information from a study of the incidence of emphysema in smokers and non-smokers.

Information	Males	Females
Number of individuals in the study	25	25
Mean age when diagnosed / years	53.1	54.2
Range of ages when diagnosed / years	32 to 77	34 to 68
Number of smokers	5	6
Number of non-smokers	20	19
Number of smokers with emphysema	1	6
Number of non-smokers with emphysema	0	0

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Criticise the design of this study.

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(Total for question = 14 marks)

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Q12.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A person with diabetes has a blood glucose level that can be too high.

When the blood glucose level of a person without diabetes becomes too high, the liver stores glucose as a polysaccharide.

(a) Which polysaccharide does the liver store?

(1)

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☒ **A** amylopectin

☒ **B** cellulose

☒ **C** glycogen

☒ **D** starch

(b) Blood glucose levels can become high following the digestion of carbohydrates.

Which of the following can be digested to release glucose?

(1)

☒ **A** both fructose and sucrose

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☒ **B** both fructose and galactose

☒ **C** both galactose and lactose

☒ **D** both lactose and sucrose

(c) Diabetes is a risk factor for cardiovascular disease.

(i) One estimate is that there are 415 million people with diabetes in the world and that 46% of these people are undiagnosed.

Calculate the number of people who have undiagnosed diabetes.

(1)

Answer .....

(ii) There are two types of diabetes, Type I and Type II.

The treatment for Type I diabetes is different from the treatment for Type II diabetes, so it is important for a correct diagnosis to be made.

Diagnosis can be difficult, particularly in people aged between 20 and 40 years old.

A genetic screening method is now available for the diagnosis of diabetes.

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Explain why doctors are more likely to screen individuals once they develop diabetes than use methods such as prenatal testing.

(3)

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(Total for question = 6 marks)

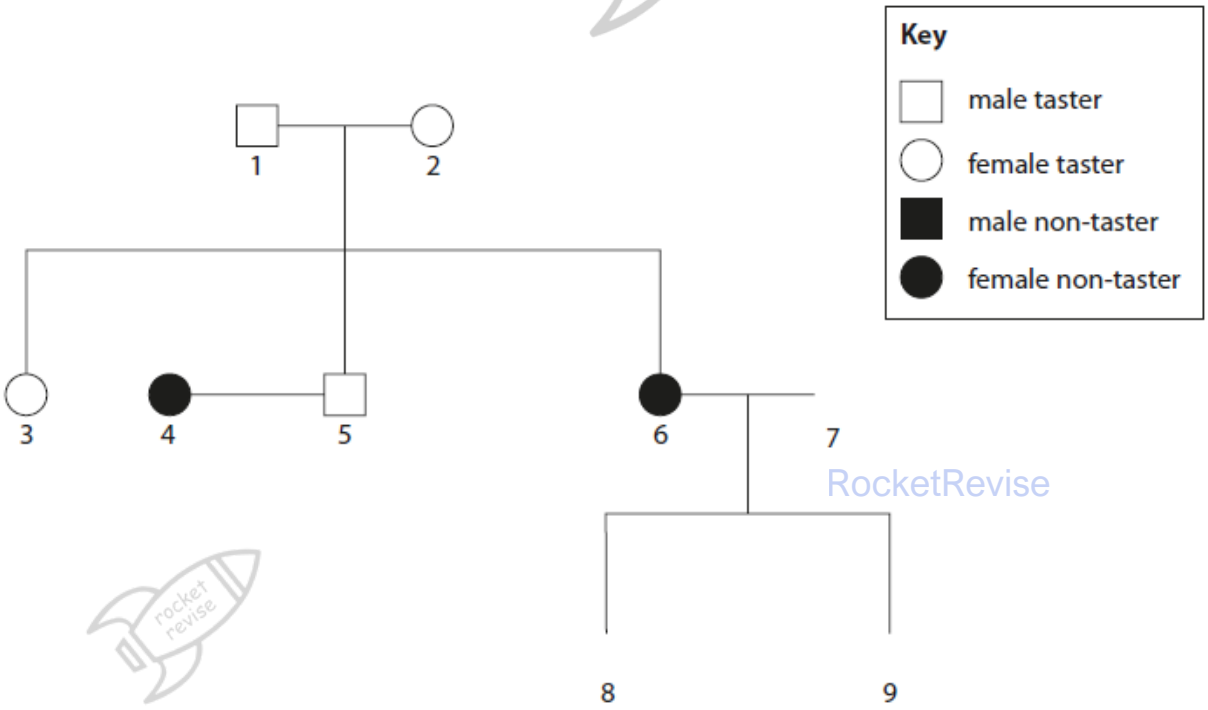
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Q13.

Phenylthiocarbamide (PTC) is a chemical that has a very bitter taste to some individuals (tasters).

The ability to taste PTC is determined by a gene that codes for a bitter-taste receptor on the tongue.

The pedigree diagram shows some of the tasters and non-tasters in a family.



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(a) Complete the diagram to show the following information:

(1)

- individual 7 as a male taster
- individual 8 as a male non-taster
- individual 9 as a female taster.

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(b) Describe the difference between each of the following pairs of terms, using the information in the pedigree diagram to illustrate your answer.

(i) Gene and allele

(2)

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(ii) Genotype and phenotype

(2)

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(c) Explain which is the dominant allele.

Use the information in the pedigree diagram to support your answer.

(2)

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(d) Explain why this gene is unlikely to be located on the X chromosome.

Use the information in the pedigree diagram to support your answer.

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(2)

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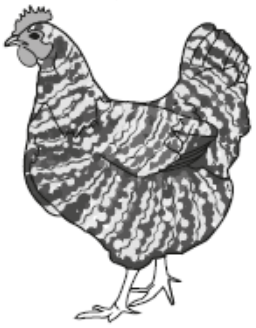
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(Total for question = 9 marks)

Q14.

The drawing shows a speckled chicken. These chickens have a mixture of black and white feathers.



The colour of the feathers of a chicken is an example of codominance.

One parent of this speckled chicken had white feathers and the other parent had black feathers.

(a) Describe the difference between each of the following pairs of terms, using feather colour to illustrate your answer.

(i) Gene and allele

(3)

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(ii) Genotype and phenotype

(3)

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(b) A black chicken was mated with a speckled chicken. They had 25 chicks.

Determine the expected number of speckled chicks.

You **must** use a genetic diagram.

(3)

Answer .....

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(c) In an experiment, several pairs of speckled chickens were mated together.

They produced 480 chicks.

The table shows the expected number of speckled chicks, white chicks and black chicks. It also shows the actual number of each type of chick.

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Steps in the calculation for the statistics test	Colour of feathers of chicks		
	Speckled	White	Black
Observed number (O)	243	125	112
Expected number (E)	240	120	120
(O-E)			
$\frac{(O-E)^2}{E}$			

This table can be used in a statistics test.

(i) Name the statistics test being used to analyse these data.

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(1)

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(ii) Complete this table to show the missing values.

(2)

(iii) Calculate $\sum \frac{(O-E)^2}{E}$

(1)

Answer .....

(iv) Explain how a critical value table could be used to accept or reject a null hypothesis for this experiment.

(2)

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(Total for question = 15 marks)

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Q15.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Humans store energy as glycogen.

(a) (i) Which is the correct statement about the formation of glycogen?

(1)

☒ **A** α glucose molecules join together by a condensation reaction

☒ **B** α glucose molecules join together by a hydrolysis reaction

☒ **C** β glucose molecules join together by a condensation reaction

☒ **D** β glucose molecules join together by a hydrolysis reaction

(ii) Name the bond that joins two glucose molecules together.

(1)

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(iii) Explain how the structure of glycogen relates to its role as an energy storage molecule.

(3)

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(b) Von Gierke disease is one type of glycogen storage disease (GSD).

(i) Between one in 20 000 and one in 25 000 babies are born with GSD.

About 25% of patients with GSD are thought to have Von Gierke disease.

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In one country, 3.8 million babies were born in one year.

Estimate the number of babies born each year with Von Gierke disease in this country.

(2)

Answer .....

(ii) Von Gierke disease is an inherited disease.

Suggest why the majority of these babies are born to parents who are not affected by Von Gierke disease.

(2)

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(Total for question = 9 marks)

Q16.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

People with lactose intolerance cannot digest lactose.

Lactose intolerance is due to a lack of the enzyme lactase.

(a) (i) Which are the products of lactose digestion?

(1)

☐ **A** fructose and galactose

☐ **B** fructose and glucose

☐ **C** galactose and glucose

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☒ **D** glucose and glucose

(ii) Explain how the three-dimensional structure of lactase affects the mechanism of action of this enzyme.

(3)

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(b) People with lactose intolerance can drink lactose-free milk.

Lactose-free milk is produced by treating milk with lactase.

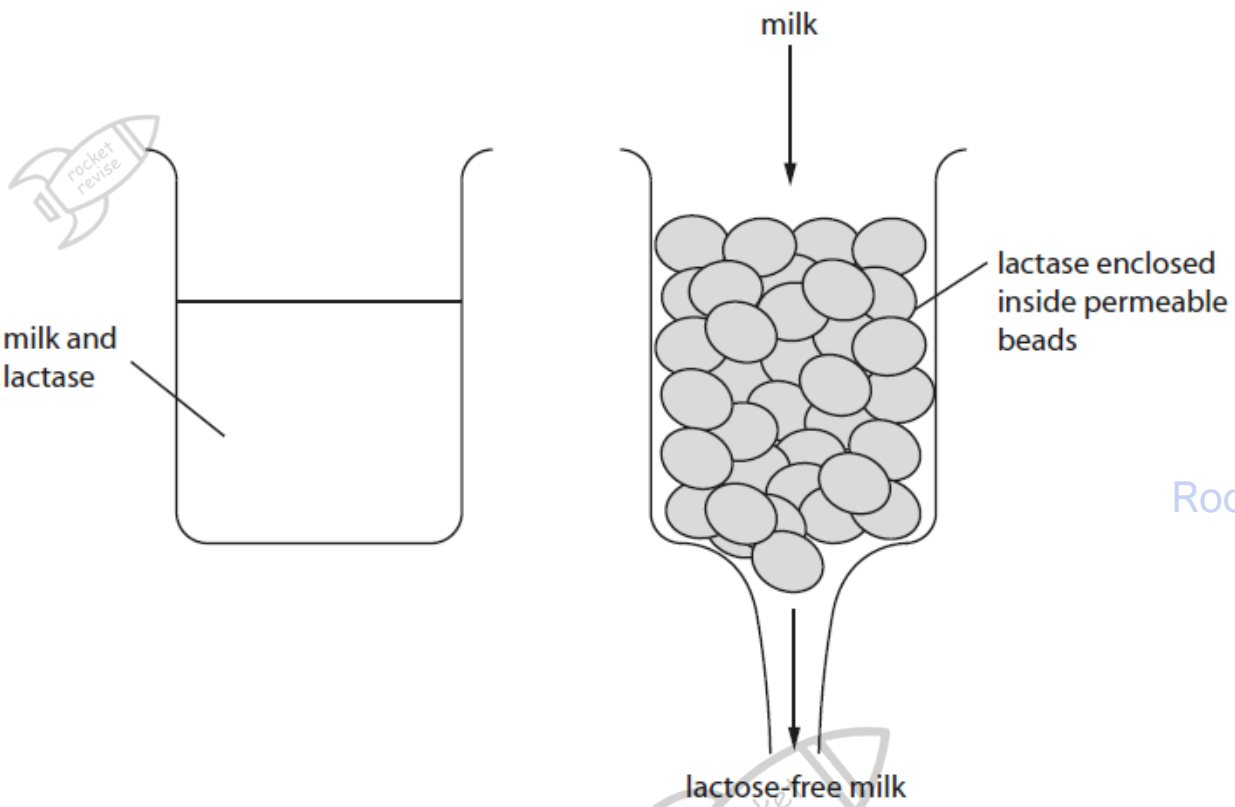
There are two ways of removing lactose from milk:

- mixing a solution of lactase with the milk (free lactase)
- enclosing the lactase inside permeable beads and pouring the milk over them (immobilised lactase).

The diagrams show these two methods.

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(i) Suggest **one** advantage of using immobilised lactase.

(1)

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(ii) The table shows the effect of pH on the activity of free lactase and immobilised lactase.

pH	Activity of free lactase / a.u.	Activity of immobilised lactase / a.u.
2	0	0
3	0	38
4	75	75
5	94	98
6	63	76
7	56	63
8	28	35

Explain the effects of pH on the activity of these two enzymes.

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(4)

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(iii) Suggest how the rate of activity of the lactase could be measured.

Include appropriate units in your answer.

(2)

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(c) Congenital lactose intolerance (CLI) is an extremely rare genetic disorder.

Most people with CLI are found in one country.

Suggest why most people with CLI are found in one country.

(2)

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(Total for question = 13 marks)

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Q17.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Mutations can give rise to cancer.

(a) What is a mutation?

(1)

☒ **A** a change in the amino acid sequence in DNA

☒ **B** a change in the amino acid sequence in a protein

☒ **C** a change in the base sequence in DNA

☒ **D** a change in the base sequence in a protein

(b) Name **two** types of mutation.

(1)

1 .....

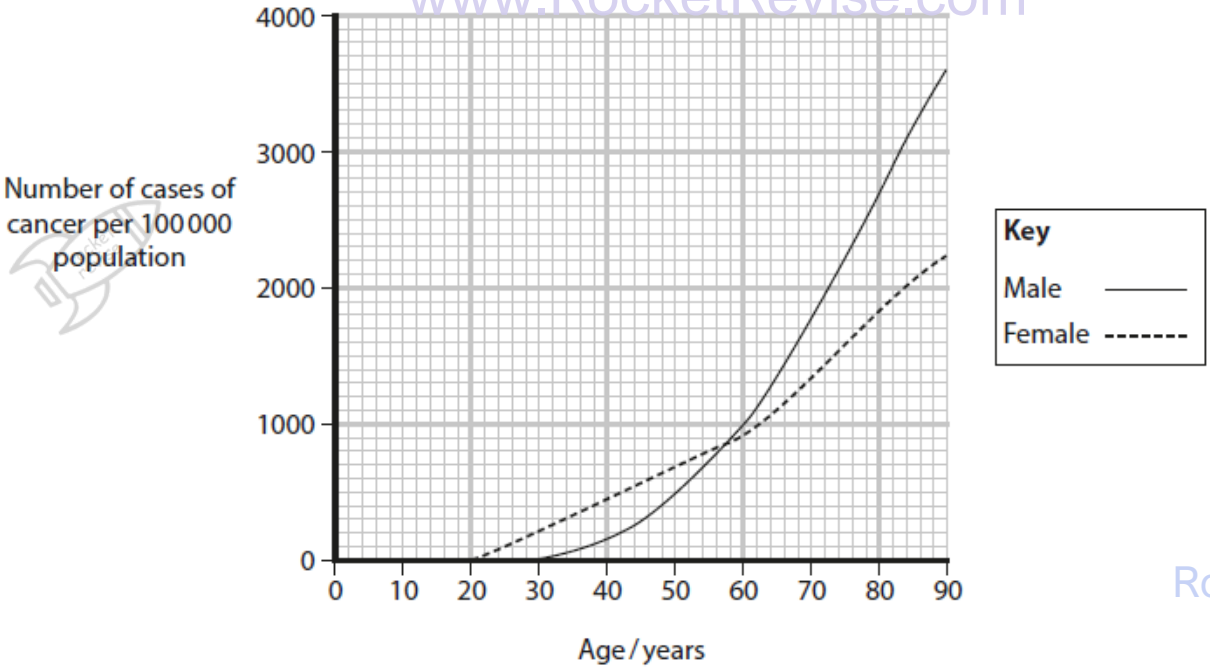
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2 .....

(c) The graph shows the number of cases of one type of cancer in a human population.

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Describe the effect of age and sex on the number of cases of cancer.

(3)

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(Total for question = 5 marks)

Q18.

Sickle cell disease is caused by a gene mutation affecting the β-globin chain of haemoglobin.

(a) The mutation occurs in the seventh triplet code of this gene.

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This mutation results in the amino acid Glu being replaced with the amino acid Val.

The table shows the sequence of bases in the first part of the DNA in a person who does not have sickle cell disease. It also shows the corresponding sequence of amino acids in the β-globin chain.

Position of triplet code	first	second	third	fourth	fifth	sixth	seventh	eighth	ninth
DNA	AUG	GUG	CAC	CUG	ACU	CCU	GAG	GAG	AAG
β-globin chain	(start)	Val	His	Leu	Thr	Pro	Glu	Glu	Lys

(i) Give the seventh triplet code in the gene for the β-globin chain in a person who has sickle cell disease.

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(1)

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(ii) Name the type of mutation that causes sickle cell disease.

(1)

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(iii) The amino acid Glu is hydrophilic (polar) and the amino acid Val is hydrophobic (non-polar).

Suggest why this mutation causes haemoglobin molecules to stick together.

(3)

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(b) In 2020, about 140 million babies were born in the world.

About 305 800 babies are born with sickle cell disease each year.

Estimate the ratio of babies born with the disease to babies not born with the disease.

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(2)

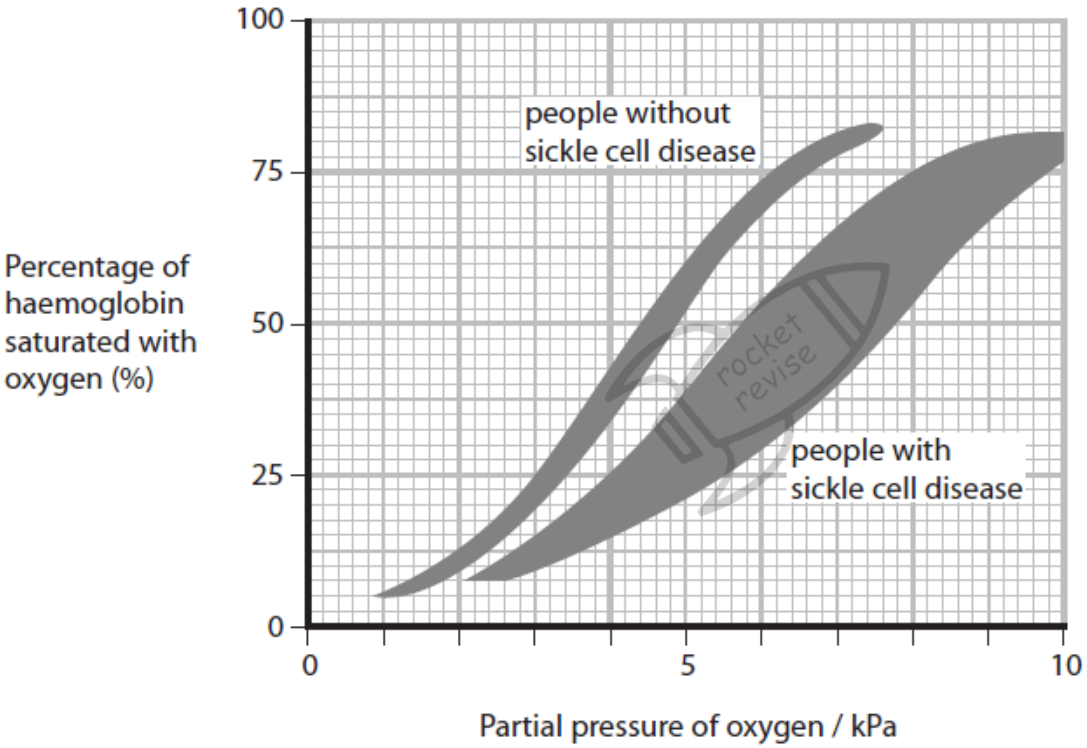
Answer .....

(c) The red blood cells in a person with sickle cell disease are sickle shaped and less elastic. They also have a shorter lifespan than healthy red blood cells.

These sickle shaped red blood cells cannot carry as much oxygen as healthy red blood cells and they get stuck in the capillaries.

The graph shows oxygen dissociation curves for groups of people with sickle cell disease and those without the disease.

The shaded areas represent the range of values for each group of people.



(i) A p50 value is the partial pressure of oxygen that results in 50% saturation of haemoglobin.

Calculate the largest difference in the p50 values between a person with sickle cell disease and a person without this disease.

Give your answer to an appropriate number of decimal places.

(2)

Answer ..... kPa

(ii) Identify **two** conclusions, other than the difference in p50 values, that can be made from this graph.

(2)

Conclusion 1

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Conclusion 2

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(iii) The lifespan of a red blood cell in a person with sickle cell disease is 11 days.

This is 9.17% of the lifespan of a healthy red blood cell.

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Calculate the lifespan of a healthy red blood cell.

Give your answer to the nearest day.

(1)

Answer ..... days

(iv) Sickle cell disease can result in death.

Explain why the changes in the structure of haemoglobin and the shape of the red blood cells could result in death in a person with sickle cell disease.

(3)

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(Total for question = 15 marks)

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Q19.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Mutations can give rise to cancer or genetic disorders.

(a) Cancer is one of the main causes of death in the world.

In 2018, in the UK, there were 541 589 deaths in total and 166 800 of these were due to cancer.

Calculate the ratio of deaths caused by cancer to deaths not caused by cancer.

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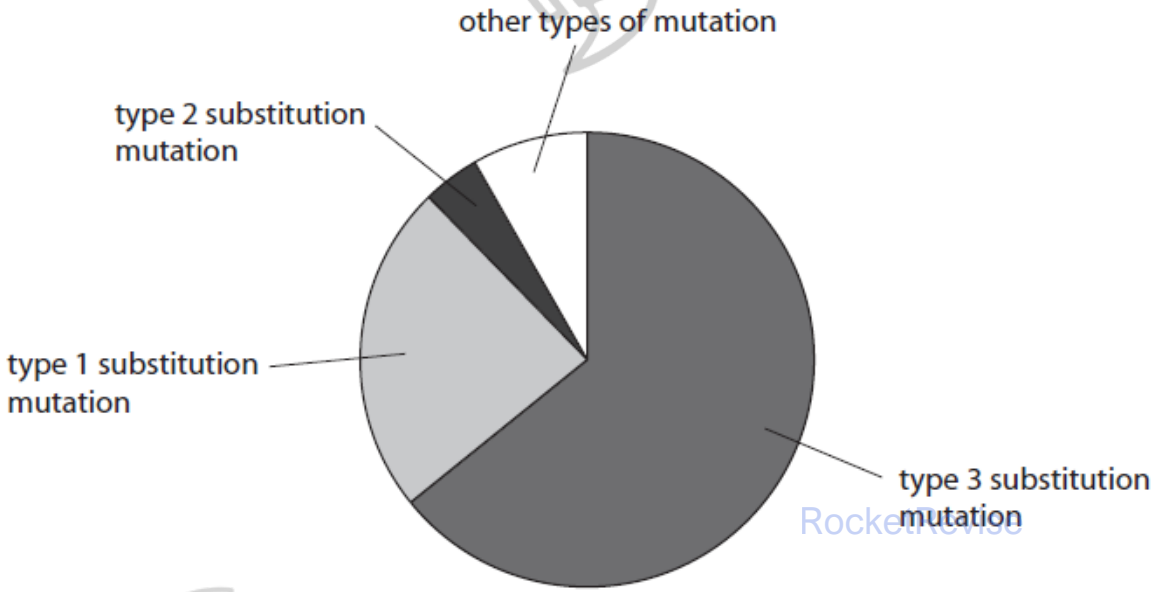
Give your answer to two decimal places.

(2)

Answer .....

(b) Scientists have identified many mutations in cancer cells and are trying to identify the mutations that are significant.

The chart shows the proportion of cancers caused by some types of mutation.



(i) Name **two** other types of mutation.

1 .....

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2 .....

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(ii) Estimate the percentage of cancers caused by type 3 substitution mutations.

(1)

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Answer

.....%

(iii) A type 1 substitution mutation in a gene alters the DNA and mRNA but does **not** affect the protein synthesised from the mutated gene.

Explain how a substitution mutation can alter the DNA and mRNA but **not** the protein.

(4)

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(c) Phenylketonuria (PKU) is a genetic disorder.

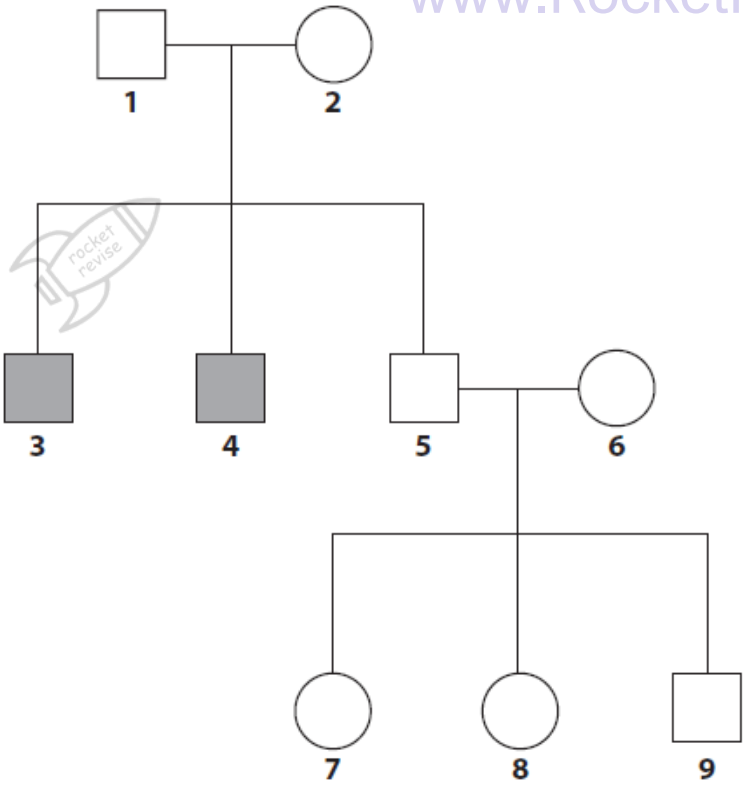
This disorder is inherited in a similar way to cystic fibrosis.

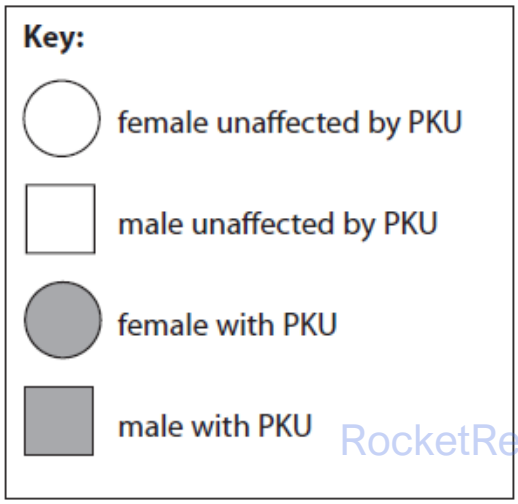
The pedigree diagram shows the phenotypes of individuals in one family affected by PKU.

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(i) What is the probability of couple **1** and **2** having a fourth child that is a male affected by PKU?

(1)

- ☒ **A** 0.000
- ☒ **B** 0.125
- ☒ **C** 0.250
- ☒ **D** 0.500

* (ii) Discuss the extent to which this pedigree diagram can be used to identify the genotypes of all the members of this family.

(6)

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(Total for question = 15 marks)

Q20.

Red-green colour blindness is an inherited condition.

The table shows some statistics on red-green colour blindness in some countries.

Country	Population size in millions	Number of people who are red-green colour blind in millions	Percentage of population who are red-green colour blind (%)
India	1380		5.2
Japan	126	3.024	2.4
Malaysia	32	1.024	
Philippines	108	3.024	2.8

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(a) Complete the table to show the number of people who are red-green colour blind in India and the percentage of people who are red-green colour blind in Malaysia.

(2)

(b) The percentage of males in Malaysia who are red-green colour blind is 6.7% and the percentage of females who are red-green colour blind is 0.4%.

(i) Calculate the ratio of red-green colour blind males to red-green colour blind females in Malaysia.

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(1)

Answer .....

(ii) Explain why the percentage of red-green colour blind males is greater than the percentage of red-green colour blind females.

(4)

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(Total for question = 7 marks)

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Q21.

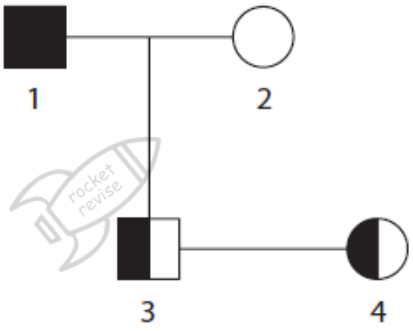
Red blood cells have a number of different groups of proteins on the cell surface membrane.

The inheritance of some of these groups of proteins is an example of codominance.

The pedigree diagram shows the blood type of individuals in part of one family. Individuals 1 and 2 are both homozygous for blood type.

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Key

- male with blood group M
- female with blood group N
- male with blood group MN
- female with blood group MN

(a) Explain why the inheritance of blood type, shown in this diagram, is an example of codominance.

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(2)

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(b) Determine the ratio of the possible blood type phenotypes of the children of individuals 3 and 4.

You must draw a genetic diagram.

(4)

Genetic diagram:

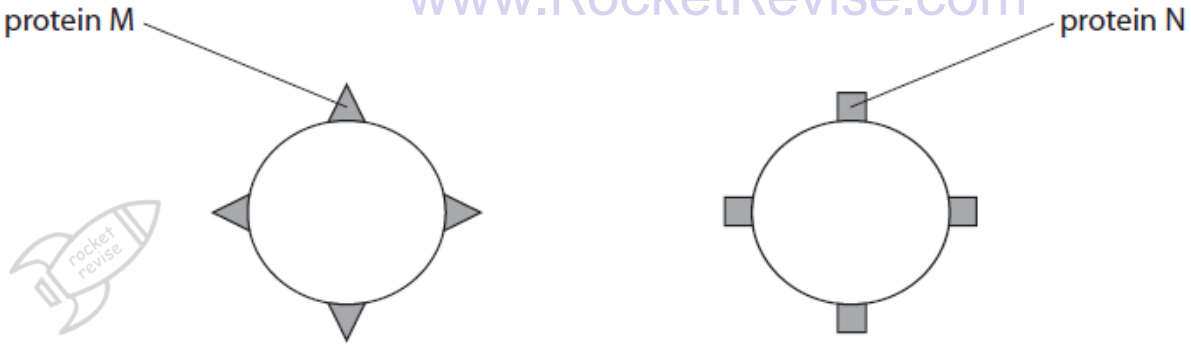
Ratio .....

(c) The gene that determines this blood type codes for proteins present on the cell membranes of red blood cells.

The diagram shows red blood cells from two members of the family shown in the pedigree diagram.

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Individual **1** Individual **2**

(i) Explain how the alleles present in individual 3 result in the proteins found on the surfaces of his red blood cells.

(3)

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* (ii) Discuss the possible effects that a mutation in one of the genes coding for this blood type could have on the phenotype of individual **3**.

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(Total for question = 15 marks)

Q22.

Cystic fibrosis results from mutations in the CFTR gene.

Mutations may occur at a number of positions within the gene, resulting in a range of clinical symptoms and severity of cystic fibrosis.

The effect of combining three different drugs to treat people with cystic fibrosis was investigated.

People with cystic fibrosis were divided into two groups. One group was given the three-drug combination and the other group was given a control treatment.

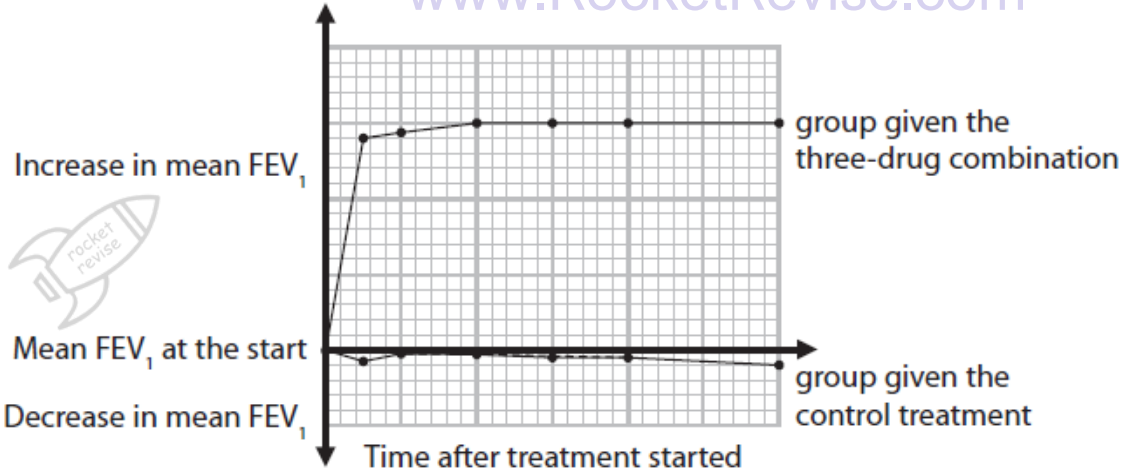
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The volume of air each person could forcefully blow out in one second (FEV_1) was measured.

The graph shows the results of this investigation.

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(a) Explain why the change in mean FEV_1 was used in this investigation.

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(3)

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(b) Explain why the results of this investigation **do not** show that this treatment is more effective than using each drug on its own.

(2)

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(c) Suggest why a combination of drugs could be more effective in treating people with cystic fibrosis than each drug used on its own.

(2)

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(Total for question = 7 marks)

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Q23.

Prenatal testing is used to screen for genetic disorders in developing embryos.

(a) Phenylketonuria (PKU) is one genetic disorder that is screened for by prenatal testing.

This disorder is inherited in a similar way to cystic fibrosis.

Explain why a healthy couple might choose to have a prenatal test for PKU.

Draw a genetic diagram in your answer.

(4)

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*(b) A group of women, who had recently given birth to healthy babies, were asked about their opinions on screening for four different genetic disorders, A, B, C and D.

They were asked three questions about each disorder:

1 Would they be screened themselves with a blood test?

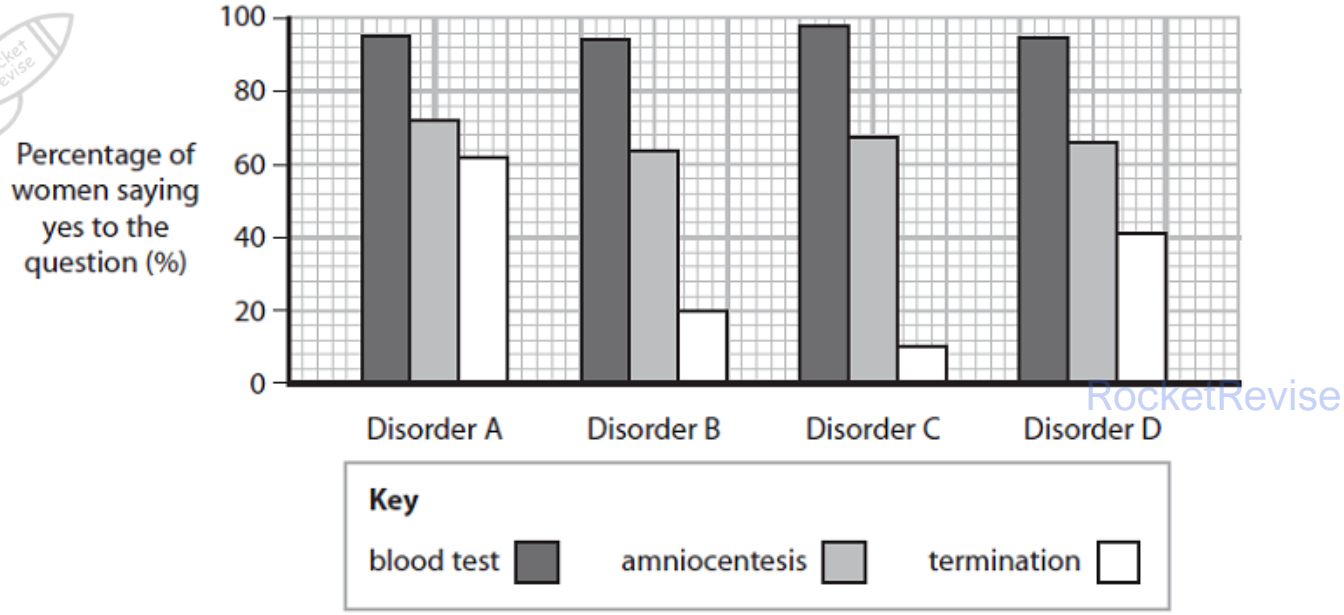
2 Would they have amniocentesis to screen for the genetic disorders?

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3 Would they have a pregnancy termination (abortion) if an unborn child was found to have the genetic disorder?

The graph shows the percentage of women who said yes to each question for each disorder.



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The table gives some information about each of these genetic disorders.

Genetic disorder	Information
A	muscles are weak and movement is difficult symptoms can be managed some types are lethal in childhood
B	treatable does not shorten life expectancy
C	treatable most individuals live into adulthood
D	intellectual disability medical care results in individuals leading healthy lives

Discuss the responses to the three questions.

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(Total for question = 10 marks)

Q24.

(a) Explain how human lungs are adapted for rapid gas exchange.

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(b) Cystic fibrosis is an inherited condition which reduces gas exchange.

The most frequently observed CFTR allele associated with cystic fibrosis carries the ΔF508 mutation.

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(i) State what is meant by the term **allele**.

(1)

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(ii) The CFTR protein coded for by this mutation has one missing amino acid compared to the functioning protein.

Explain how this mutation results in a non-functioning CFTR protein.

(4)

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(iii) Explain why people with cystic fibrosis may develop lung infections.

(4)

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(Total for question = 12 marks)

Q25.

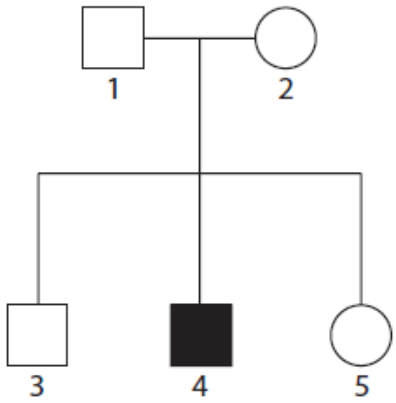
Answer the question with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

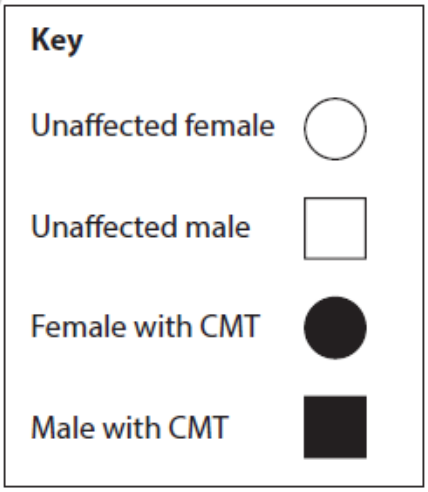
Charcot-Marie-Tooth (CMT) is an inherited disease that results in the gradual loss of function and sensation in the hands, arms, legs and feet.

Over 100 different gene mutations have been identified that cause CMT.

(a) The pedigree diagram shows the inheritance of one form of CMT in part of a family.

The allele for CMT can be represented by the letter c in this **recessive** pattern of inheritance.





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(i) Which row of the table shows the genotype and phenotype for individual 1?

(1)

		genotype	phenotype
☒	**A**	CC	unaffected homozygous
☒	**B**	Cc	unaffected heterozygous
☒	**C**	unaffected heterozygous	CC
☒	**D**	unaffected homozygous	Cc

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(ii) Which allele combinations could individual 3 have?

(1)

☒ **A** CC only

☒ **B** cc only

☒ **C** CC or cc

☒ **D** CC or Cc

(iii) What is the probability that a fourth child would be a female with CMT?

(1)

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☒ **A** 1 in 2

☒ **B** 1 in 4

☒ **C** 1 in 8

☒ **D** 1 in 16

(b) Other forms of CMT show a dominant pattern of inheritance.

(i) In the USA, 5% of people with CMT have one of these dominant forms of the disease.

In the USA, there are a total of 126 000 people with all forms of CMT.

Calculate the number of people in the USA who have one of these dominant forms of CMT.

(1)

Answer .....

(ii) An individual with a dominant form of CMT will have the allele combination TT or Tt.

Draw a genetic diagram to show the possible allele combinations of the children of the following parents:

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- parent 1 does not have CMT
- parent 2 has the allele combination Tt

Indicate whether each child will have CMT or be unaffected.

(3)

(c) Another form of CMT can be inherited as a sex-linked trait on the X chromosome.

Explain why a male with CMT **cannot** pass the disease on to his **sons**.

(2)

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(Total for question = 9 marks)

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Q26.

Cystic fibrosis is an inherited disorder caused by a mutation in the gene coding for the CFTR protein.

(a) Explain how this mutation results in reduced gas exchange in the lungs.

(3)

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(b) The number of new cases of cystic fibrosis in the USA in 2015 was 964.

It is predicted that this value will have fallen to 839 by 2025.

(i) Calculate the predicted percentage decrease in the number of new cases from 2015 to 2025.

(1)

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Answer ..... %

(ii) Explain why the number of cases could fall.

(3)

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*(c) One complication of cystic fibrosis is lung disease caused by the build-up of bacteria in the airways.

This may be due to the inability of the person to clear mucus from the airways.

People with cystic fibrosis can inhale a salt solution to treat their symptoms. This gives improvements in the clearance of mucus, lung function and quality of life, for a short time.

It has been suggested that pre-treatment with drug A could extend the effects of inhalation of a salt solution.

Devise an investigation to determine if pre-treatment with drug A is more effective than inhalation of a salt solution on its own.

Give reasons for your choice of method.

(6)

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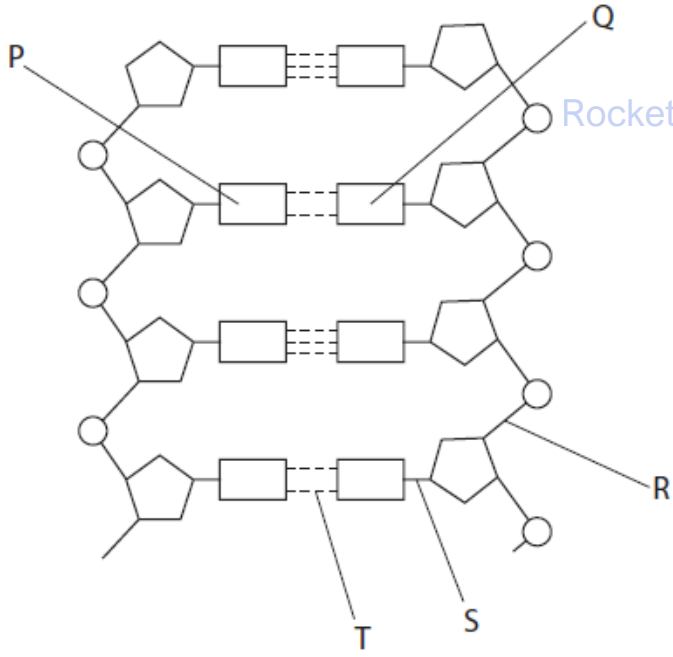
(Total for question = 13 marks)

Q27.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Errors in DNA replication can give rise to mutations.

(a) The diagram shows part of a DNA molecule.



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(i) The base labelled **Q** is thymine.

Which is the base labelled **P**?

(1)

☒ **A** adenine

☒ **B** cytosine

☒ **C** guanine

☒ **D** uracil

(ii) Which row of the table identifies the bonds labelled **R**, **S** and **T**?

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(1)

	R	S	T
☒ **A**	covalent	phosphodiester	hydrogen
☒ **B**	hydrogen	covalent	phosphodiester
☒ **C**	hydrogen	phosphodiester	covalent
☒ **D**	phosphodiester	covalent	hydrogen

(b) Name **two** types of gene mutation.

(1)

1 .....

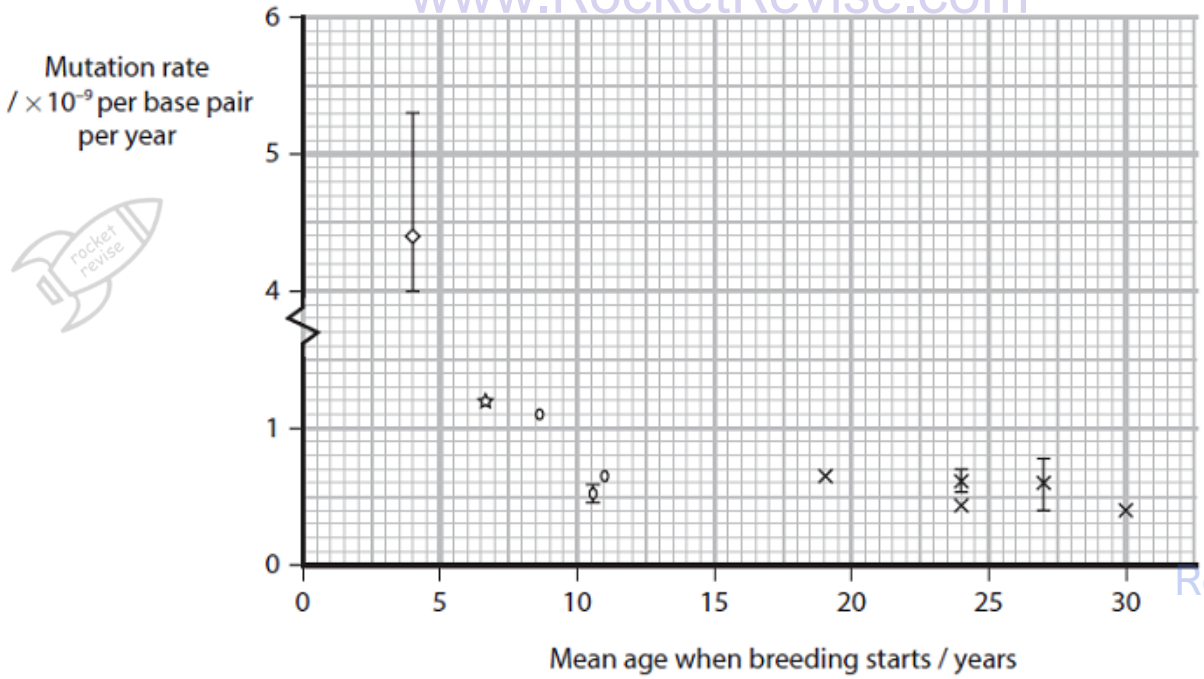
2 .....

(c) The graph shows the estimated mutation rate in four groups of primates.

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Describe **two** conclusions that can be made about the mutation rate in these four groups of primates.

(2)

1 .....

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2 .....

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(Total for question = 5 marks)

Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Haemophilia is a group of rare conditions that affects the ability of the blood to clot.

These conditions are caused by mutations in the genes coding for factors involved in the blood clotting process.

(a) Haemophilia is inherited in a similar way to red-green colour blindness and most people who have it are male.

Explain why most people who have haemophilia are male.

(4)

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(b) Two types of haemophilia are haemophilia A and haemophilia B.

In 2021, there were 7.87×10^9 people in the world.

There were 185 218 people with haemophilia A and 37 998 people with haemophilia B.

(i) Calculate the ratio of people with haemophilia A to those with haemophilia B. Give your answer to **two** significant figures.

(2)

Ratio ..... :1

(ii) Which is the percentage of the world population who had haemophilia A, in 2021?

(1)

☒ **A** 2.35×10^5

☒ **B** 2.35×10^3

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☒ **C** 2.35×10^{-3}

☒ **D** 2.35×10^{-5}

(iii) The number of people with haemophilia is only an estimate.

Give **one** reason why this is only an estimate.

(1)

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(c) In haemophilia, activation of the blood clotting pathway results in less thrombin being produced than in healthy individuals.

Explain how reduced thrombin will affect blood clotting.

(3)

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(Total for question = 11 marks)

Q29.

The types of lipid consumed in the diet can affect the health of individuals.

(a) Tropical sprue is a disorder that limits the absorption of nutrients into the blood.

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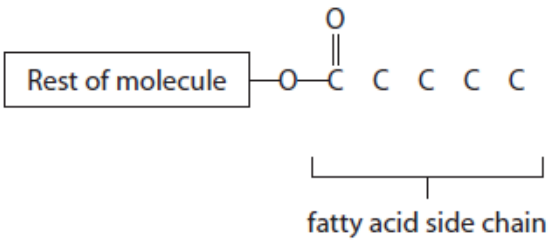
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Consumption of unsaturated lipids has been proposed as a cause of tropical sprue.

The diagram shows part of an unsaturated lipid.

Complete the diagram to show one possible structure of a fatty acid side chain of an unsaturated lipid.

(2)



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(b) Consumption of high levels of cholesterol is associated with atherosclerosis.

In the development of atherosclerosis, lesions form in the coronary arteries and these develop into atheromas (plaques).

In an investigation, one group of primates was fed a diet supplemented with cholesterol and saturated fatty acids.

A second group was fed a diet supplemented with cholesterol and unsaturated fatty acids.

The primates were fed these diets from a young age and then monitored over a period of 60 months.

Primates are a higher order of mammals that include monkeys.

The results of the investigation are shown in the graphs.

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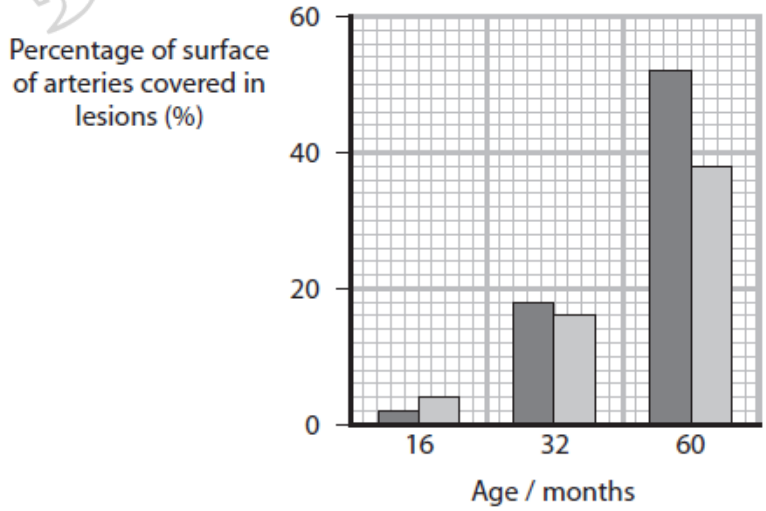
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Key

saturated lipids

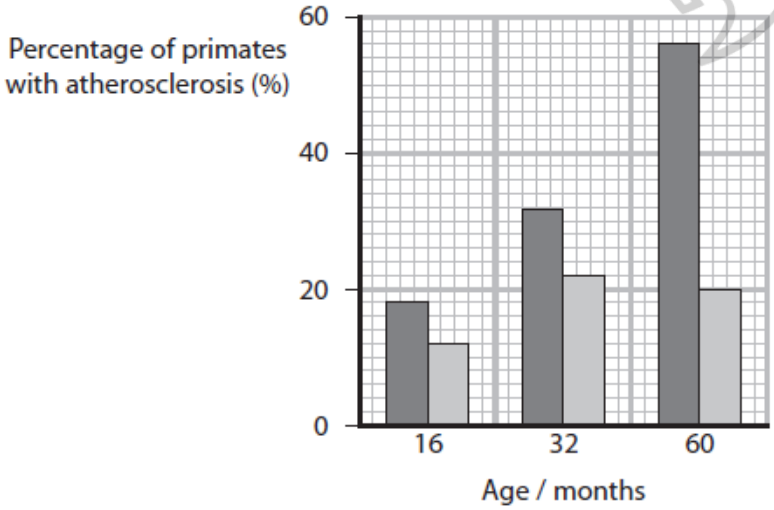
unsaturated lipids

Graph 1



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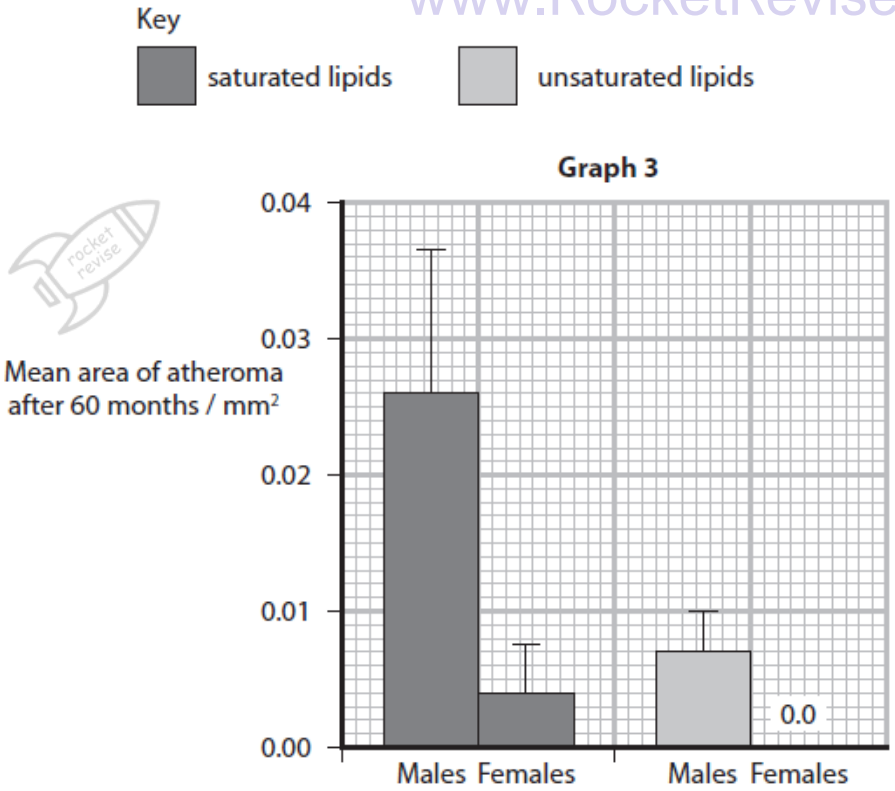
Graph 2



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(i) Calculate the mean rate of increase in the surface of arteries covered in lesions from 16 to 60 months in the primates given saturated lipids (graph 1).

(1)

Answer ..... percentage per month

(ii) Calculate the ratio of primates with atherosclerosis given saturated lipids to those given unsaturated lipids, at 32 months (graph 2).

(1)

Answer .....

(iii) Calculate the difference in the area of the atheromas in males compared with females, in primates given saturated fats (graph 3).

Give your answer in μm^2.

(2)

Answer .....μm^2

*(iv) These primates were fed a diet supplemented with cholesterol and either saturated lipids or unsaturated lipids.

Deduce the effects of these two diets on the development of lesions in the coronary arteries, atherosclerosis and atheromas.

(6)

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(v) Discuss the ethical issues relating to the use of primates and other animals in investigations such as this one.

(3)

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(Total for question = 15 marks)

Q30.

Scientists investigated the numbers and types of mutations in different species of animal.

Samples of tissue were collected from living animals of each species and the types of mutation determined.

(a) The types of mutation were classed as either substitution mutations or insertion and deletion mutations.

The table shows the results of this investigation.

Species of animal	Number of animals sampled	Range of number of mutations per cell	
		Substitution	Insertion and deletion
Naked mole rat	3	100 to 850	10 to 50
Mouse	10	300 to 2000	50 to 400
Dog	3	1000 to 3000	50 to 300
Human	8	200 to 350	10 to 200

(i) Describe how these types of mutation affect the **DNA base sequence** of a gene.

(2)

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(ii) Explain why these animals can survive with a greater number of substitution mutations than insertion and
deletion mutations.

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(4)

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(iii) Suggest which set of data is the most valid.

Give a reason for your answer.

(1)

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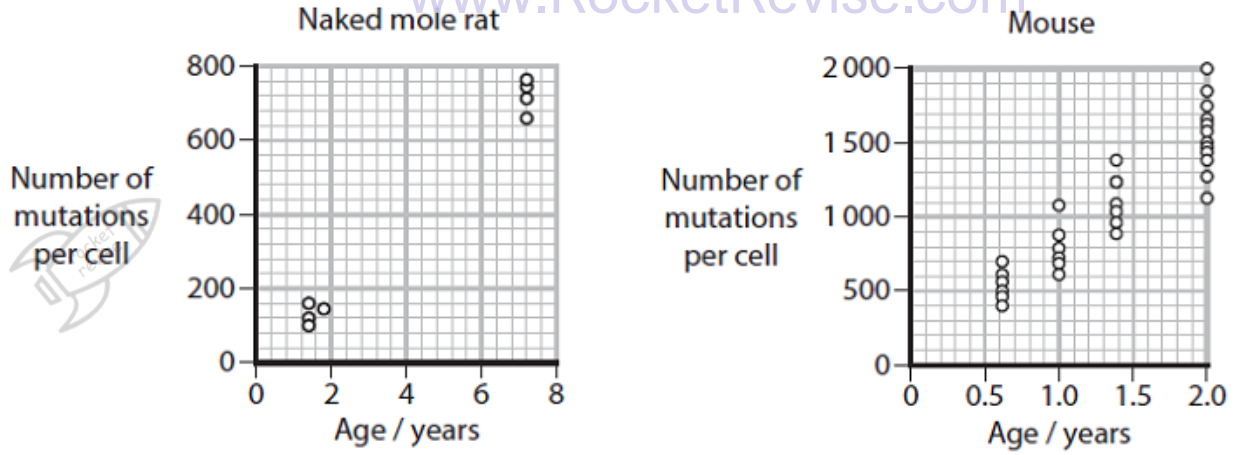
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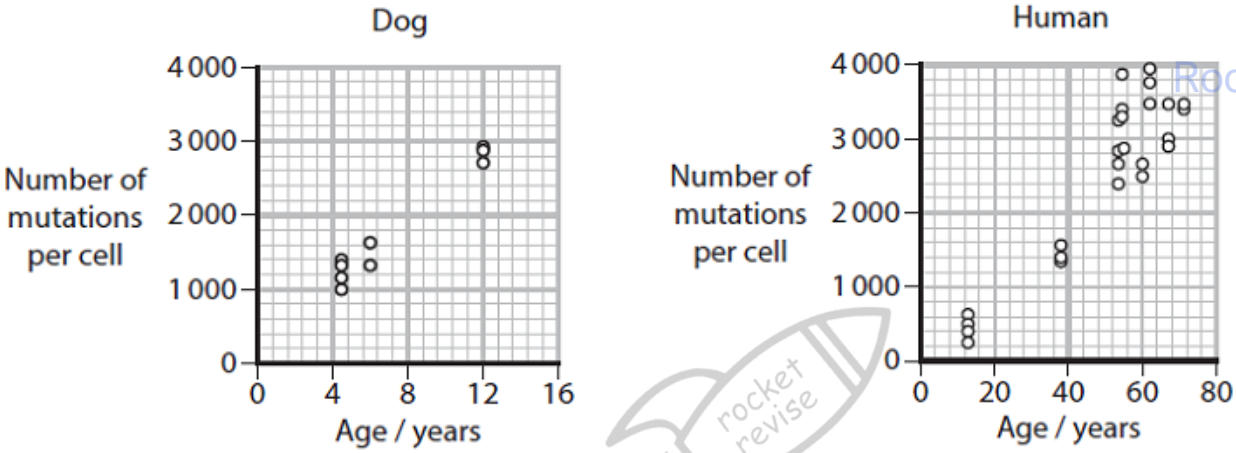
(b) The graphs show the number of mutations per cell during the lifespans of these four animals.

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(i) Calculate the mean mutation rate for the mouse.

Draw a line of best fit onto the graph to help you.

(2)

Answer ..... number of mutations per cell year^{-1}

(ii) Describe **three** conclusions that can be made from these graphs.

(3)

1 .....

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2 .....

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3 .....

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(iii) Suggest **two** reasons for the differences in the number of mutations per cell of these species of animal.

(2)

1 .....

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2 .....

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(Total for question = 14 marks)

Q31.

Warfarin is used as an anticoagulant to treat some patients.

Patients respond differently to this drug. This makes it difficult for doctors to select a safe but effective dose of warfarin to give to each patient.

Factors that affect the response of a patient to warfarin include age, body mass, liver function and genetic makeup.

(a) Explain why a safe but effective dose of warfarin needs to be given to a patient.

(2)

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(b) There are two genes that code for proteins that affect warfarin and the blood clotting process.

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Gene M:

- codes for enzyme M that breaks down warfarin into inactive molecules
- has many alleles and two of these code for much slower breakdown of warfarin.

Gene V:

- codes for enzyme V that converts vitamin K into an active form that triggers the blood clotting process
- one version of this gene results in a decreased production of the messenger RNA (mRNA) needed for the synthesis of enzyme V.

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(i) Describe the role of mRNA in the production of enzymes.

(2)

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*(ii) Pharmacogenomics analyses the genetic profile of individuals.

This analysis is used to determine the appropriate doses of a drug for individual patients.

Discuss the possible use of pharmacogenomics in selecting appropriate doses of warfarin for individual patients.

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Use the information in the question and your own knowledge of the blood clotting process and genetic screening to support your answer.

(6)

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(Total for question = 10 marks)

Q32.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Cancer can develop in many parts of the body.

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(a) State what is meant by the term **cancer**.

(2)

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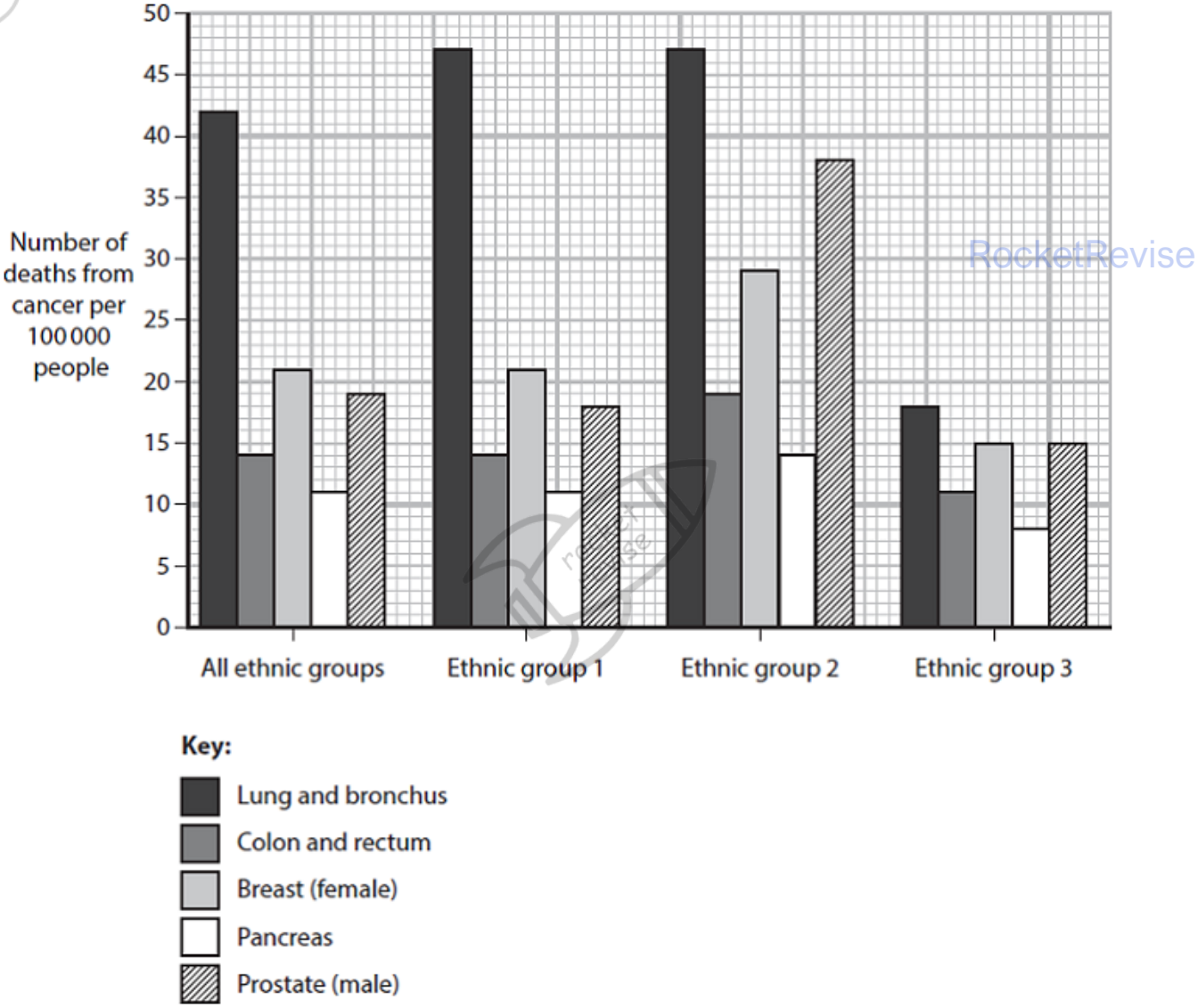
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(b) In a study, scientists determined the number of deaths caused by five types of cancer, in one country.

The scientists then analysed the number of deaths caused by these cancers in three ethnic groups.

The graph shows the results of this study.



(i) Compare and contrast the number of deaths caused by these cancers in ethnic groups 1 and 2.

(3)

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(ii) Which is the percentage difference between the number of deaths caused by lung and bronchus cancer in ethnic group 1 and ethnic group 3?

Use the formula:

$$\text{Percentage difference} = \left(\frac{\text{difference between values}}{\text{mean of the values}}\right) \times 100$$

(1)

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☒ **A** 0.45%

☒ **B** 0.89%

☒ **C** 44.62%

☒ **D** 89.23%

(iii) Calculate how many more males died of prostate cancer in ethnic group 2 than in ethnic group 1.

(1)

Answer .....

(iv) Comment on how useful it is to determine the number of deaths from these causes of cancer in the 'all ethnic groups' part of this graph.

(2)

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(Total for question = 9 marks)

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Q33.

Osteogenesis imperfecta (OI) is a group of rare genetic disorders which results in bones that are not formed properly and can break easily.

There are several types of OI, as a number of different genes can be affected.

Some of the symptoms of OI can be life-threatening.

The commonest types of OI are caused by mutations in genes coding for the structure of collagen.

(a) In some types of OI, the collagen is poorly formed and lacks strength.

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Explain how a substitution mutation could result in collagen that lacks strength.

(4)

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(b) In most types of OI the allele responsible for causing the condition is **dominant**.

Determine the probability that a child could inherit OI if one parent has OI and is heterozygous and the other parent does not have OI and is therefore homozygous recessive.

Use a genetic diagram to support your answer, indicating the phenotypes of the offspring and their corresponding genotypes.

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(4)

Genotype of parent with OI: .....

Genotype of parent without OI: .....

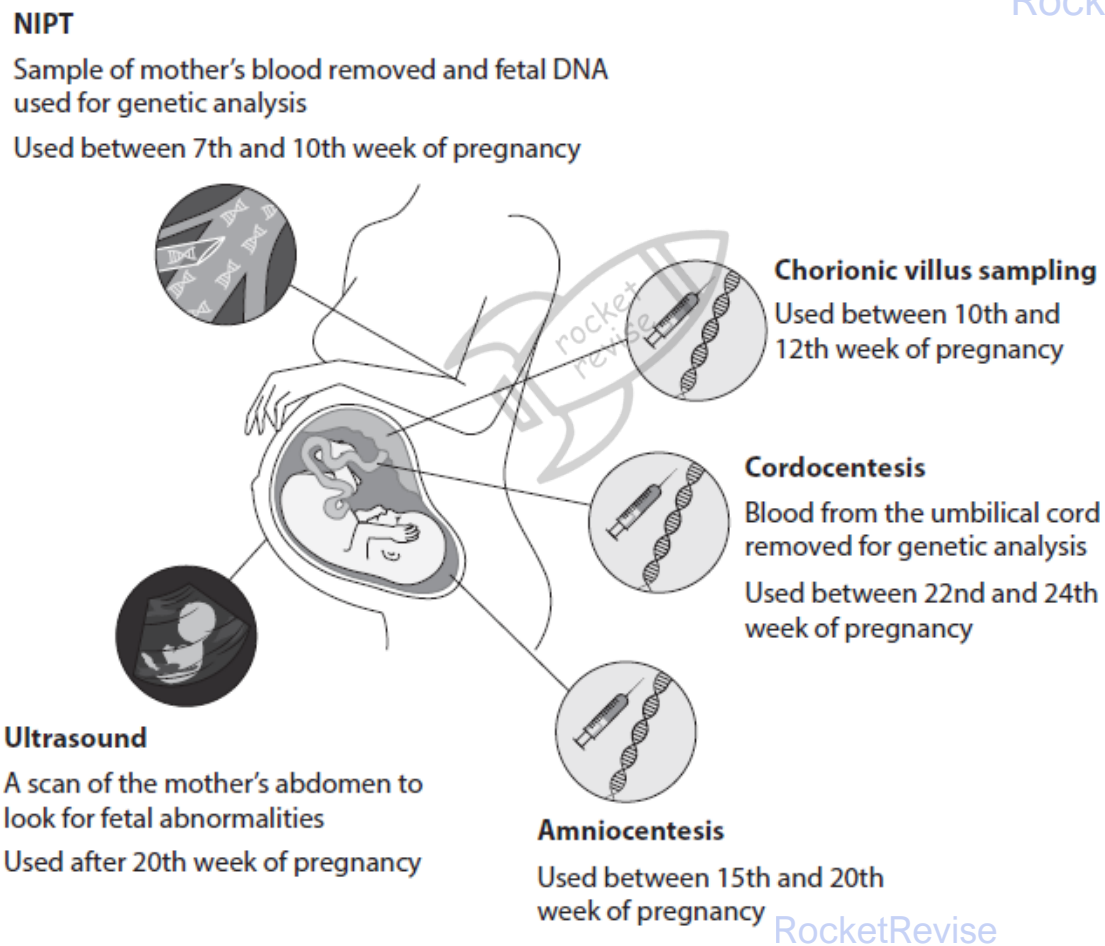
Probability that a child could inherit OI .....

*(c) Genetic screening can be used by couples who are at risk of having a child with OI.

The genetic screening options include pre-implantation genetic diagnosis (PGD) and prenatal testing.

The diagram gives some information about some of the types of prenatal testing available to these couples.

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Discuss the advantages and disadvantages of the genetic screening options available to couples at risk of having a child with OI .

Use all the information in the question and your own knowledge to support your answer.

(6)

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(Total for question = 14 marks)

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